Unit 12, Lesson 5: Volume of Right Circular Cylinders – Part 2

Benchmark

MA.7.GR.2.3 Solve mathematical and real-world problems involving volume of right circular cylinders.

Learning Target

- [ ] I can find the volume of a right circular cylinder.

12.5.1 Warm-Up: What Do You Wonder?

1. A silo is a cylindrical container that is used on farms to hold large amounts of goods, such as grain. The artist Cam Scale painted a mural across 7 adjacent silos. The painting is of a young girl standing in an open meadow looking up into the open sky. The mural took about 4 weeks to paint.

   a. Describe at least two steps you think the artist needed to take before painting this mural.

   b. What is something you wonder about this artist's project?

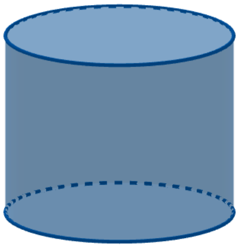
12.5.2 Guided Instruction: Finding the Volume of a Right Circular Cylinder

1. Complete the statements.

To find the volume of a cylinder, [○ add / ○ multiply] the [○ area / ○ circumference] of the base of the cylinder [○ to / ○ by] the height of the cylinder.

2. Landon says he can use the formula $V = \pi \left(\frac{d}{2}\right)^2 h$ to find the volume of a cylinder. Shayna disagrees and says the formula for finding the volume of a cylinder is $V = \pi r^2 h$. Discuss with your partner who you think is correct.

3. The cylinder shown has a diameter of 6 meters (m.) and a height of 4.8 m.



a. With your partner, decide who is partner A and who is partner B. Then calculate the volume of the cylinder using the formula indicated. Leave your answer in terms of π.

Partner A **Use Landon's formula,** $\boldsymbol{V = \pi \left(\frac{d}{2}\right)^2 h}$	**Partner B** **Use Shayna's formula,** $\boldsymbol{V = \pi r^2 h}$

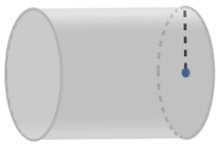
b. Both formulas should result in the same answer. Check your answer with your partner. Then, discuss the similarities and differences in your calculations. Write a summary of your discussion.

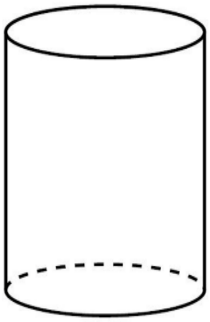
c. Complete the statement.

The volume of the cylinder is ______ π
- ○ m.
- ○ m^2.
- ○ m^3.

4. Four cylinders are shown and specific measurements are given. The measurements include the radius, r, diameter, d, height, h, or circumference, C.

a. For each cylinder, select which formula you would use to find the volume.

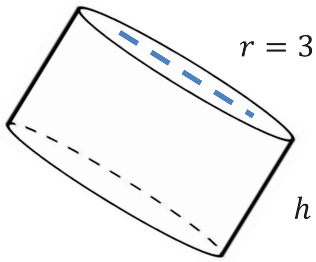
Cylinder	**Formula**
 $r = 3$ in. $h = 5$ in.	○ $V = \pi\left(\frac{d}{2}\right)^2 h$ ○ $V = \pi r^2 h$
$d = 3$ cm.  $h = 11$ cm.	○ $V = \pi\left(\frac{d}{2}\right)^2 h$ ○ $V = \pi r^2 h$
$h = 4.1$  $r = 2$ m.	○ $V = \pi\left(\frac{d}{2}\right)^2 h$ ○ $V = \pi r^2 h$

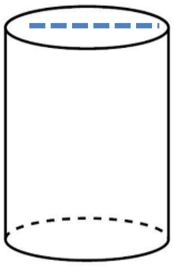
 $C = 6\pi$ units $h = 10$ units	○ $V = \pi\left(\frac{d}{2}\right)^2 h$ ○ $V = \pi r^2 h$

b. Either formula can be used to calculate the volume of a cylinder. Compare your selections above with your partner. Discuss your reasoning for each selection.

12.5.3 Guided Practice: Using Diameter or Radius to Find the Volume of a Right Circular Cylinder

1. Calculate the volume of each right circular cylinder. Leave your answers in terms of π.





2. On Mrs. Scarham's farm, there is a silo with a height of 28 feet (ft.) and diameter of 12 ft.

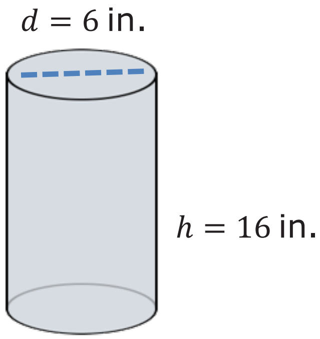
   a. Draw a sketch of the silo and label the dimensions.

   b. How many cubic feet of grain can this silo hold? Use 3.14 for π.

12.5.4 Collaboration: Error Analysis

1. Simone calculated the volume of a cylinder with a diameter of 6 inches (in.) and height of 16 in. Her calculations are shown.

$$(\pi \cdot 6^2)(16)$$
$$36\pi \cdot 16$$
$$576\pi \text{ in}^2$$



Simone made errors in her work. Work with your partner to determine what errors were made. Write a note to Simone identifying her errors and explaining how she can correct them.

12.5.5 Your Turn!

1. The dimensions of four cylinders are given in the column on the left. Match each set of dimensions with the corresponding volume of the cylinder.

1. ____ $r = 2$ units; $h = 5$ units	A. 36π units³
2. ____ $r = 9$ units; $h = 3$ units	B. 98π units³
3. ____ $d = 6$ units; $h = 4$ units	C. 20π units³
4. ____ $d = 7$ units; $h = 8$ units	D. approximately 763 units³

2. Find the volume of the cylinder shown. Use $\frac{22}{7}$ for π. Round your answer to the nearest thousandth.

$h = 5$ units

$r = 2$ units

3. The figure shows two grain storage silos. The diameter of each measures 24 feet (ft.) and the height of each cylinder measures 51 ft. Find the volume of the cylindrical portion of one of the silos. Use 3.14 for π.

12.5.6 Wrap-Up: Ticket Out the Door

1. At a farm, animals are fed from buckets of grain. Each bucket of grain is a cylinder with a diameter of 12 inches. The height of the bucket is 14 inches. What is the volume of grain the bucket can hold? Use $\frac{22}{7}$ for π.